# The Sharp Monotone Euclidean Extension Constant

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## Source Question and Status

Gargiulo and Ok define the order-preserving extension constants $e_{k,\uparrow}(X,Y)$ and $e_{\uparrow}(X,Y)$ in their concluding section. For $\mathbb{R}^n$ with Euclidean metric and coordinatewise order, they obtain the coordinatewise upper bound $\sqrt{n}$ and ask whether it is best; they also ask for finite-$k$ estimates [1, p. 12].

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> have $e_{\uparrow}(X,Y) > 1$. In fact, our proof of this result shows that $e_{2,\uparrow}(X,Y) > 1$ in this case. It should thus be quite interesting to seek upper bounds for the quantity $e_{k,\uparrow}(X,Y)$ when $X$ and $Y$ are arbitrary Hilbert posets (with $\dim X \geq 2$), but this will very much depend on the structure of the partial orders of $X$ and $Y$. For instance, as $e_{\uparrow}(X,\mathbb{R}) = 1$, we have $e_{\uparrow}(X,\mathbb{R}^n) \leq \sqrt{n}$ for all $n \in \mathbb{N}$ (where $X$ is any metric poset), but we do not presently know if this is the best upper bound, nor have good estimates for $e_{\uparrow,k}(X,\mathbb{R}^n)$.

This packet proves that $\sqrt{n}$ is exactly sharp even for finite radial metric posets, and that the constant has already saturated when the datum has $2n$ points. It also records a necessary scope correction: the scalar identity used to deduce the upper bound requires the order of $X$ to be radial. Without radiality there is no finite dimension-only bound. Status: *candidate full solution, likely valid*, subject to human review.

## Definitions and the Coordinatewise Upper Bound

For nonempty $S \subset X$, let $e_{\uparrow}(X,S,Y)$ be the least $K$ such that every order-preserving 1-Lipschitz $f : S \to Y$ has an order-preserving $K$-Lipschitz extension to $X$, and put

$$e_{k,\uparrow}(X,Y) = \sup_{0<|S|\leq k} e_{\uparrow}(X,S,Y), \qquad e_{\uparrow}(X,Y) = e_{\infty,\uparrow}(X,Y).$$

The scalar monotone extension theorem says that $e_{\uparrow}(X,\mathbb{R}) = 1$ exactly when the order on $X$ is radial [1, Sec. 2.2]; see also [2]. Hence, if $X$ is radial and $f : S \to \mathbb{R}^n$ is order-preserving and 1-Lipschitz, each coordinate of $f$ has an order-preserving 1-Lipschitz extension $F_j : X \to \mathbb{R}$. Then

$$F = (F_1, \ldots, F_n), \qquad \|F(x) - F(y)\|_2 \leq \sqrt{n}\, d_X(x,y), \tag{1}$$

so $e_{\uparrow}(X,\mathbb{R}^n) \leq \sqrt{n}$ for every radial $X$.

## Sharpness and Finite-Set Saturation

**Theorem 1.** *For every $n \ge 1$ there is a finite radial metric poset $X_n$ such that*

$$e_{2n,\uparrow}(X_n, \mathbb{R}^n) = e_\uparrow(X_n, \mathbb{R}^n) = \sqrt{n}.$$

*More generally, for $1 \le m \le n$,*

$$e_{2m,\uparrow}(X_n, \mathbb{R}^n) \ge \sqrt{m}.$$

*Consequently, the $\sqrt{n}$ upper bound in (1) is best possible.*

*Proof.* Let

$$X_n = \{\top, \bot\} \cup \{p_j^+, p_j^- : 1 \le j \le n\}.$$

Give $X_n$ the discrete metric: distinct points have distance one. Order it as a height-three poset, with

$$\top \succ p_j^\pm \succ \bot \quad (1 \le j \le n),$$

and no comparabilities between distinct middle points. Every partial order on a discrete metric space is radial: whenever either radial inequality has a nontrivial premise, the two distances being compared are both one.

Let $S_n = \{p_j^\pm : 1 \le j \le n\}$ and, with $(e_j)$ the standard basis of $\mathbb{R}^n$, define

$$f(p_j^+) = \tfrac{1}{2} e_j, \qquad f(p_j^-) = -\tfrac{1}{2} e_j. \tag{2}$$

The set $S_n$ is an antichain, so $f$ is order-preserving. It is 1-Lipschitz: opposite points on the same axis have image distance one, and all other distinct image pairs have distance $1/\sqrt{2}$ or less.

Let $F : X_n \to \mathbb{R}^n$ be any order-preserving extension. For each $j$, monotonicity and (2) give

$$F(\top)_j \ge \tfrac{1}{2}, \qquad F(\bot)_j \le -\tfrac{1}{2}.$$

Thus every coordinate of $F(\top) - F(\bot)$ is at least one, and

$$\|F(\top) - F(\bot)\|_2 \ge \sqrt{n}.$$

Since $d_{X_n}(\top, \bot) = 1$, every extension has Lipschitz constant at least $\sqrt{n}$. Hence $e_{2n,\uparrow}(X_n, \mathbb{R}^n) \ge \sqrt{n}$. The reverse inequality follows from radiality of $X_n$ and (1), proving both equalities.

For $m \le n$, use only the $2m$ data points $p_j^\pm$, $1 \le j \le m$, with the same formula. The first $m$ coordinates of the top-bottom difference are forced to be at least one, giving the lower bound $\sqrt{m}$. $\square$

In particular, for every $k \ge 2n$, $e_{k,\uparrow}(X_n, \mathbb{R}^n) = \sqrt{n}$. The construction supplies the explicit finite-$k$ lower estimate

$$e_{k,\uparrow}(X_n, \mathbb{R}^n) \ge \sqrt{\min\{n, \lfloor k/2 \rfloor\}} \qquad (k \ge 2).$$

## Why Radiality Cannot Be Dropped

The source paragraph says parenthetically that $X$ is "any metric poset," but its premise $e_\uparrow(X, \mathbb{R}) = 1$ is equivalent to radiality. The following shows that this is not merely a technical distinction.

**Proposition 1.** *For every $n \ge 1$ there is a metric poset $X$ with*

$$e_{2,\uparrow}(X, \mathbb{R}^n) = \infty.$$

*Thus no bound depending only on $n$ holds for arbitrary metric posets.*

*Proof.* For $0 < \varepsilon < 1$, let $X_\varepsilon = \{a, b, c\}$, with

$$d(a,b) = d(a,c) = 1, \qquad d(b,c) = \varepsilon,$$

and with $c \succ a$ as its only strict order relation. On $S = \{a, b\}$ put $f(a) = e_1$ and $f(b) = 0$. The two points of $S$ are incomparable and their distance is one, so $f$ is order-preserving and 1-Lipschitz. Every order-preserving extension satisfies $F(c)_1 \geq 1$, hence

$$\mathrm{Lip}(F) \geq \frac{\|F(c) - F(b)\|_2}{d(c,b)} \geq \frac{1}{\varepsilon}. \tag{3}$$

Now take the disjoint union of $X_{1/r}$, $r = 2, 3, \ldots$, retain the order inside each component, and put distance two between distinct components. This is a metric poset (indeed, its topology is discrete). Applying (3) in its $r$th component shows that its two-point extension constant is infinite. □

Proposition 1 corrects only the scope of the sentence; it does not weaken Theorem 1, whose extremal domains are radial.

## Novelty Check and Verification

The cheap run indexes contained no answer. Bounded searches through 17 August 2026 used the exact source sentence, title, arXiv id, and the terms *monotone Lipschitz extension constant*, *radial metric poset*, and *coordinatewise Euclidean order*. They found arXiv v1 and the scalar predecessor but no later solution or correction. This is not a priority claim.

The companion script checks all pairwise image distances, the forced coordinate gaps for $1 \leq n \leq 128$, every restricted $m$ construction, and 1000 nonradial blow-up scales. These checks are auxiliary; the proof is the finite argument above.

## Human Review Recommendation

Verify that the intended upper bound assumes radiality, and check the two order inequalities forcing the images of $\top$ and $\bot$. The latter gives a particularly short audit of sharpness.

## References


[1] E. Gargiulo and E. A. Ok, *Order-Preserving Extensions of Hadamard Space-Valued Lipschitz Maps*, arXiv:2603.03549v1 (2026).

[2] E. A. Ok, *Order-preserving extensions of Lipschitz maps*, J. Aust. Math. Soc. **118** (2025), 91–107; arXiv:2305.00511.